

PVsyst - Simulation report

Grid-Connected System

Project: LEARNING

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 3300 Wp

KOTDWARLālpur - India



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PVsyst V8.0.21

VC0, Simulation date:
17/03/26 20:44
with V8.0.21

Project summary

Geographical Site

KOTDWARLālpur

India

Situation

Latitude 29.75 °(N)

Longitude 78.51 °(E)

Altitude 392 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

Lālpur

Meteonorm 8.2 (1991-2000), Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1

Fixed plane

Tilt/Azimuth 30 / 0 °

Near Shadings

no Shadings

User's needs

Monthly values

System information

PV Array

Nb. of modules 6 units

Pnom total 3300 Wp

Inverters

Nb. of units 1 unit

Total power 3 kWac

Pnom ratio 1.10

Results summary

Produced Energy	4793.6 kWh/year	Specific production	1453 kWh/kWp/year	Perf. Ratio PR	83.77 %
Used Energy	1581.7 kWh/year			Solar Fraction SF	44.13 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1

Fixed plane

Tilt/Azimuth 30 / 0 °

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Monthly values

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.	Year	
134	121	134	130	134	130	134	134	130	134	130	134	1582	kWh/mth

PV Array Characteristics

PV module

Manufacturer Generic

Model ASM-M10-144-550

(Original PVsyst database)

Unit Nom. Power 550 Wp

Number of PV modules 6 units

Nominal (STC) 3300 Wp

Modules 1 strings x 6 In series

At operating cond. (50°C)

Pmpp 2997 Wp

U mpp 225 V

I mpp 13 A

Total PV power

Nominal (STC) 3.30 kWp

Total 6 modules

Module area 15.5 m²

Cell area 14.3 m²

Inverter

Manufacturer Generic

Model MIN 3000TL-X

(Original PVsyst database)

Unit Nom. Power 3.00 kWac

Number of inverters 1 unit

Total power 3.0 kWac

Operating voltage 80-500 V

Pnom ratio (DC:AC) 1.10

Power sharing within this inverter

Total inverter power

Total power 3 kWac

Number of inverters 1 unit

Pnom ratio 1.10

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const) 20.0 W/m²K

Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 284 mΩ

Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.38 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.15 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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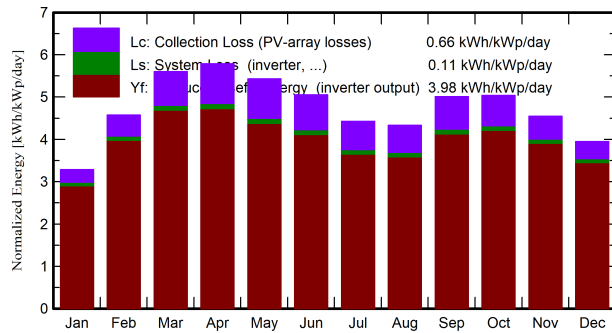
Main results

System Production

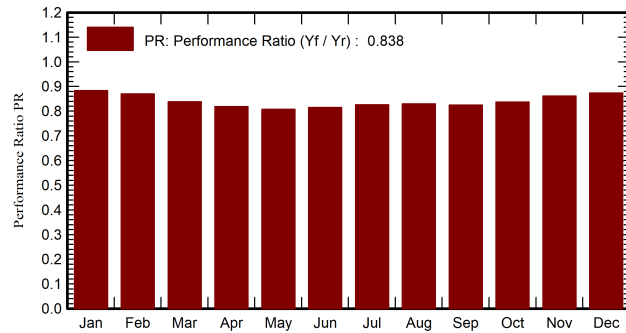
Produced Energy 4793.6 kWh/year
Used Energy 1581.7 kWh/year

Specific production 1453 kWh/kWp/year
Perf. Ratio PR 83.77 %
Solar Fraction SF 44.13 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January	77.4	44.26	12.43	102.0	100.3	306.1	134.3	47.6	249.8	86.8
February	102.1	53.63	15.81	128.2	126.0	377.5	121.3	49.6	318.3	71.8
March	151.7	66.97	21.07	173.7	170.5	492.4	134.3	60.7	419.5	73.7
April	169.8	83.30	26.28	173.5	169.9	480.7	130.0	62.1	406.5	67.9
May	179.9	95.16	29.86	168.3	164.2	460.8	134.3	67.0	381.6	67.3
June	166.8	97.33	29.69	151.6	147.9	419.4	130.0	65.8	342.1	64.2
July	148.8	95.07	28.35	137.1	133.7	385.0	134.3	66.2	307.8	68.1
August	138.9	89.80	27.52	134.3	131.0	378.4	134.3	63.7	303.9	70.7
September	138.9	71.13	26.29	150.4	147.2	420.3	130.0	58.5	350.5	71.5
October	129.3	64.23	23.54	156.2	153.6	442.6	134.3	55.7	375.9	78.6
November	100.1	47.28	18.09	136.3	134.1	397.4	130.0	52.4	335.2	77.6
December	85.5	39.21	13.81	122.5	120.7	362.9	134.3	48.7	304.7	85.6
Year	1589.1	847.38	22.76	1734.0	1699.0	4923.7	1581.7	698.0	4095.7	883.7

Legends

GlobHor Global horizontal irradiation
DiffHor Horizontal diffuse irradiation
T_Amb Ambient Temperature
GlobInc Global incident in coll. plane
GlobEff Effective Global, corr. for IAM and shadings

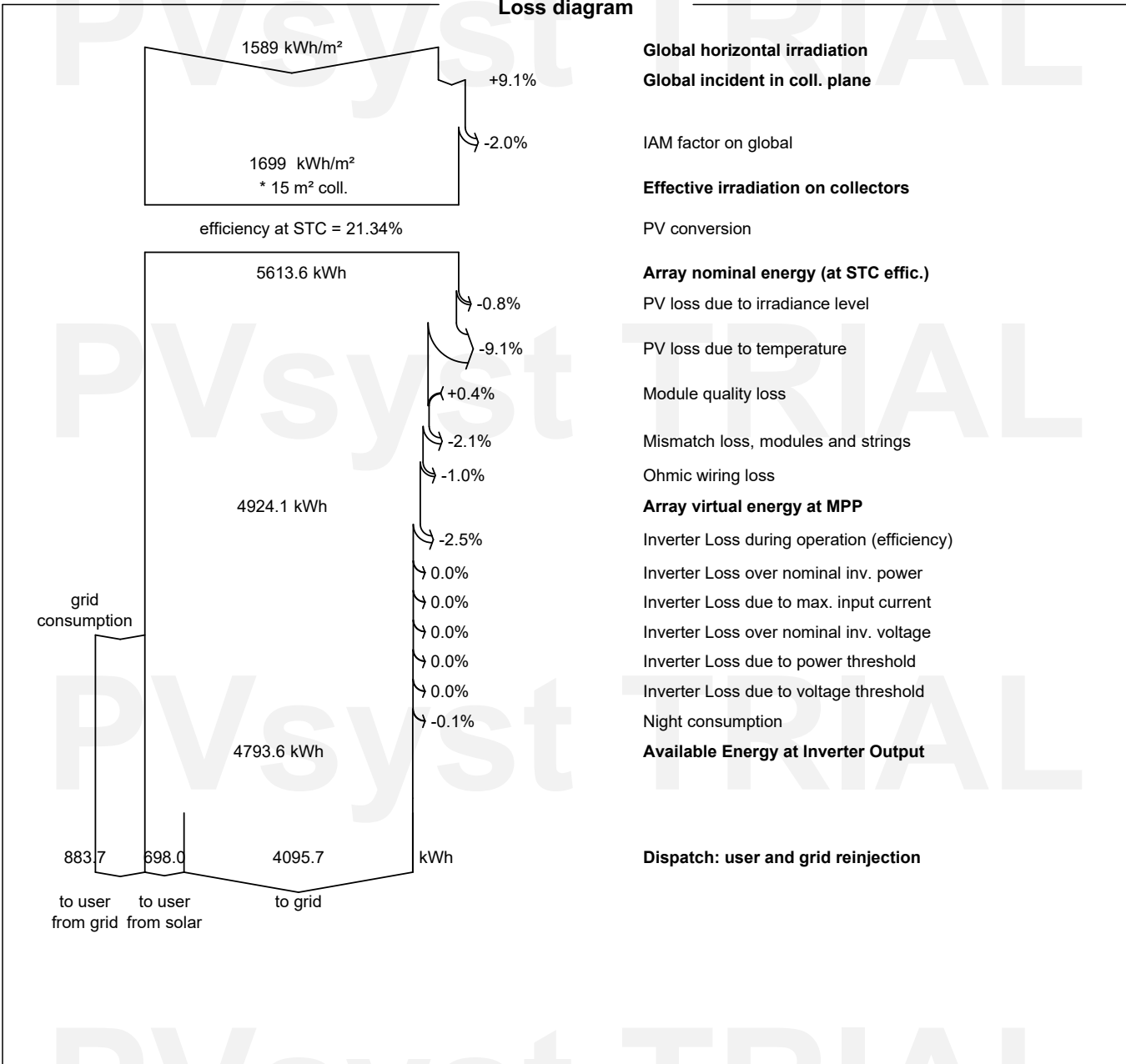
EArray Effective energy at the output of the array
E_User Energy supplied to the user
E_Solar Energy from the sun
E_Grid Energy injected into grid
EFrGrid Energy from the grid



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Loss diagram



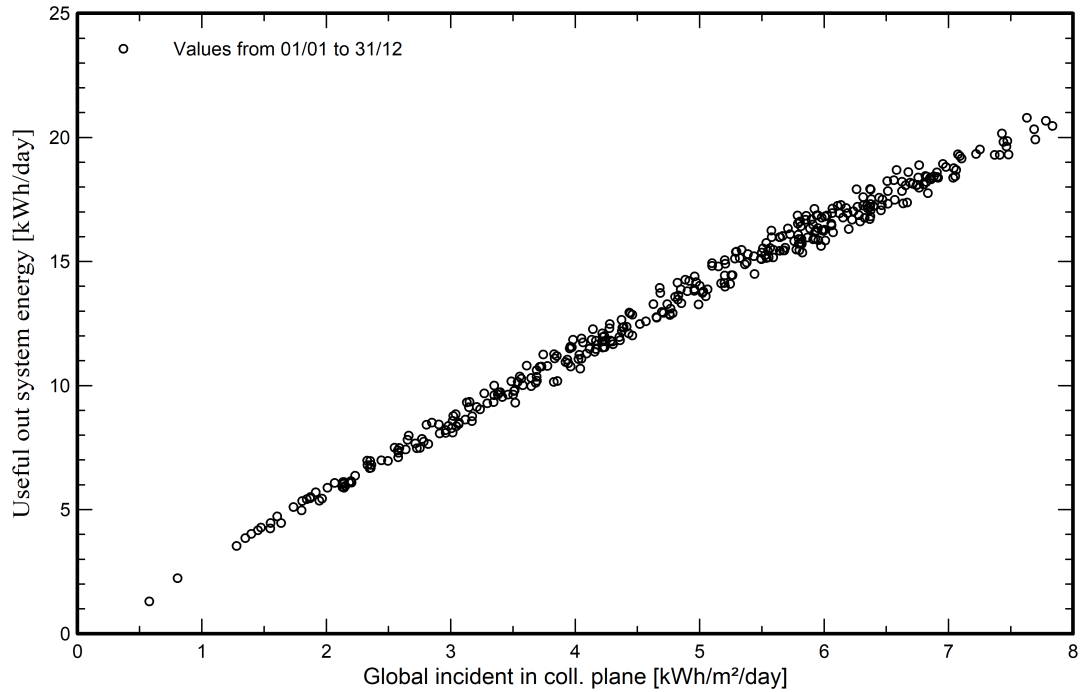


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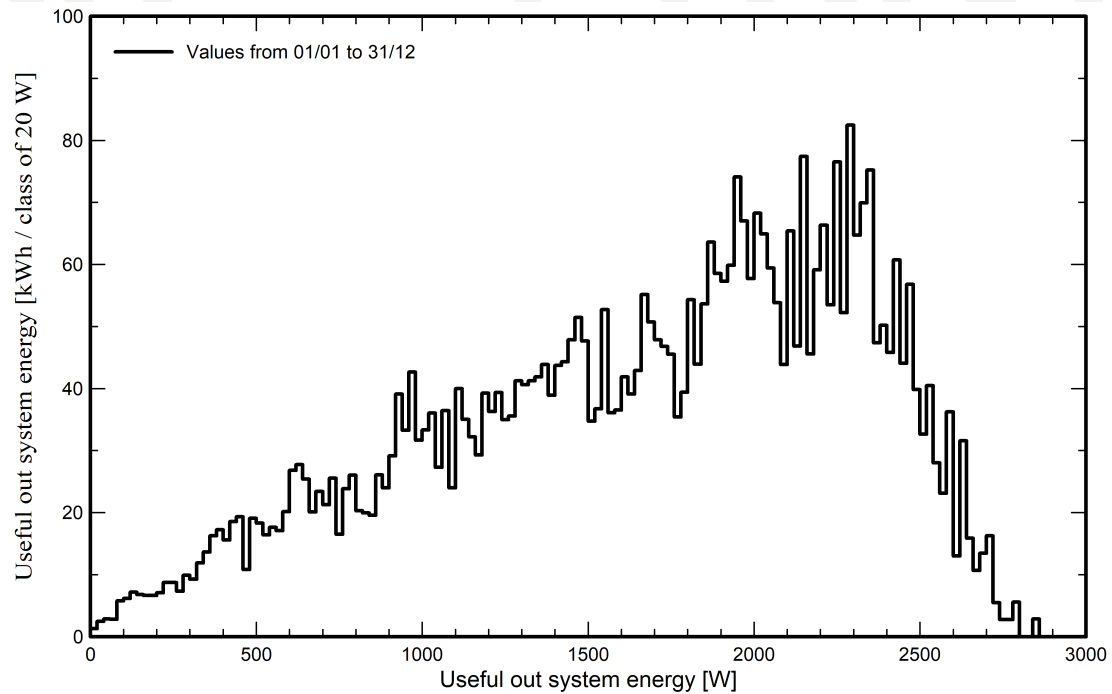
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

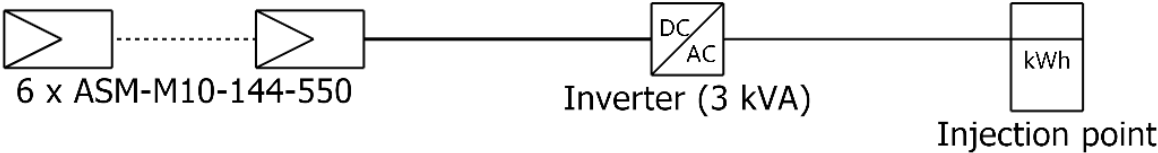




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Single-line diagram



PV module	ASM-M10-144-550
Inverter	MIN 3000TL-X
String	6 x ASM-M10-144-550

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